Harshal Sawatkar

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with 2+ years of experience leading engineering teams and building production-grade, cloud-native systems, autonomous multi-agent workflows, and real-time distributed platforms using Python, Go, Node.js, and React. Proven track record leading institutional FinTech (StockGenius AI) and enterprise data quality platforms (DataGlow) at Aavil Inc., scaling high-throughput CRMs handling 10,000+ monthly leads, and architecting LangGraph state machines, real-time WebSocket fanout gateways, and neural voice systems. Experienced across the full lifecycle from system design and schema modeling to CI/CD and high-availability Linux deployments.

TECHNICAL SKILLS

Languages: Python, Go (Golang), JavaScript (ES6+), TypeScript, SQL, C/C++

AI & Agentic Systems: LangGraph, LangChain, Multi-Agent Swarms, Azure OpenAI, NVIDIA NIM, RAG Pipelines, FAISS, Vector DBs, Azure Neural Voice, Ollama, Prompt Engineering

Backend & Distributed Systems: FastAPI, Node.js, Express.js, Go Microservices, WebSockets, Socket.IO, Server-Sent Events (SSE), RESTful APIs, Redis (Cache & CDC Queues), Upstox API Gateway

Databases & Lakehouse: PostgreSQL, MongoDB, Databricks Delta Lake, Snowflake, Redis, MySQL, SQLite, Neo4j, Prisma ORM

Frontend & Mobile: React.js (18/19), Next.js (SSR/SSG), TypeScript, React Native, Capacitor, PWA, Tailwind CSS, TanStack Virtual, Three.js

Cloud, DevOps & Infrastructure: Azure (VMs, DNS Zones, Blob Storage), AWS (EC2, S3, CloudFront), Docker, Nginx (Reverse Proxy & SSL Conf), CI/CD (GitHub Actions), Linux/Unix (Ubuntu), Systemd, Git, Postman, Jest

PROFESSIONAL EXPERIENCE

Full Stack AI Engineer (Project Lead) 2025 – Present

Aavil Inc. (aavilabs.com)

- Spearhead engineering and system architecture for flagship institutional AI & FinTech platforms: **StockGenius AI** (stockgenius.ai) and **DataGlow** (dataglow.ai), directing technical deliverables and microservice designs.
- Architected StockGenius AI's real-time streaming bridge delivering sub-50ms tick-to-client updates over WebSockets and reference-counted broker subscription pooling, slashing upstream rate-limit overhead by 85%.
- Built a concurrency-safe paper trading terminal with transactional ledger execution, eliminating race conditions during peak equity market volatility across NSE/BSE and NYSE/NASDAQ feeds.
- Engineered DataGlow's deterministic 3-Layer Scoring Engine, Meridian predictive drift forecasting (OLS regression), and CDC self-healing architecture across Databricks Delta Lake, cutting incident MTTR from hours to seconds.
- Orchestrated multi-agent LangGraph consensus committees with strict mathematical guardrails to eliminate LLM hallucinations in enterprise data quality metrics.

Full-Stack Developer Jan 2026 – Mar 2026

Zizbey Consultancy Private Limited

- Architected and delivered scalable microservices backends in Go and Node.js, handling real-time data streaming via WebSockets and secure multi-tenant RBAC authorization flows.
- Designed distributed RESTful API systems for authentication, third-party integrations, and core business logic focused on horizontal scalability under burst loads.
- Owned end-to-end product lifecycle: requirement scoping, database schema design, Go/Node.js implementation, automated testing, and CI/CD deployment on Linux environments.

Technical Head – Full Stack Development Aug 2024 – Dec 2025

REBORN SKIN INTERNATIONAL CLINICS PVT LTD (LinkedIn)

- Headed full-stack technical architecture for clinic operations, scaling an enterprise CRM platform processing 10,000+ monthly leads across 8 branches with 50+ concurrent users.
- Implemented real-time data pipelines using Redis pub/sub and WebSocket session management, reducing lead ingestion and triage latency from 2–3 minutes to sub-second response times.
- Automated lead ingestion pipelines integrating Meta Ads and Google Ads APIs with webhook validation, exponential backoff retry queues, and multi-branch attribution rules, reducing lead leakage by 40%.
- Built and maintained automated CI/CD pipelines via GitHub Actions and Docker on Linux servers, ensuring 99.9% operational uptime.

KEY PROJECTS

Ghost Agent — Autonomous Multi-Agent Outbound Sales Pipeline GitHub

Python, LangGraph, NVIDIA NIM, FastAPI, Playwright, Azure OpenAI

- Built an autonomous agentic outbound sales engine that extracts DOM tech stack signals, analyzes prospect pain points, and synthesizes hyper-personalized high-converting pitch proposals.
- Designed cyclic state machines in LangGraph with human-in-the-loop approval checkpoints, multi-turn qualification scoring, and automated pitch deck generation, reducing SDR manual qualification cycles by 70%.

ACE (Life OS) — Autonomous Life Operating System with Neural Voice Live App — GitHub
Node.js, Express.js, Azure Neural Voice, Gemini API, SQLite Vector, SSE, PWA

- Engineered a mobile-first PWA personal executive companion ('Aura') featuring conversational neural voice streaming, multi-device Server-Sent Events (SSE) synchronization, and proactive coaching telemetry.
- Integrated SQLite vector memory with direct agentic tool calling, enabling the assistant to autonomously adapt workout splits, progressive overload calculations, and calendar schedules in real time.

StockGenius AI — Institutional FinTech Terminal & AI Copilot stockgenius.ai
React 18, Node.js, FastAPI, Socket.IO, Redis, Azure OpenAI, Upstox API, TanStack Virtual, PWA

- Built 'Ask Genius' (autonomous tool-calling financial copilot), sub-50ms WebSocket streaming gateway, and virtual paper trading terminal with concurrency-safe ledgers, packaged as installable PWA and Android applications.

EDUCATION

Bachelor of Technology in Computer Science & Engineering 2020 – 2024
Priyadarshini Bhagwati College of Engineering, Nagpur

- Leadership: President – Student Placement Cell (led recruitment drives placing 100+ students)